



EXAM PAPERS PRACTICE

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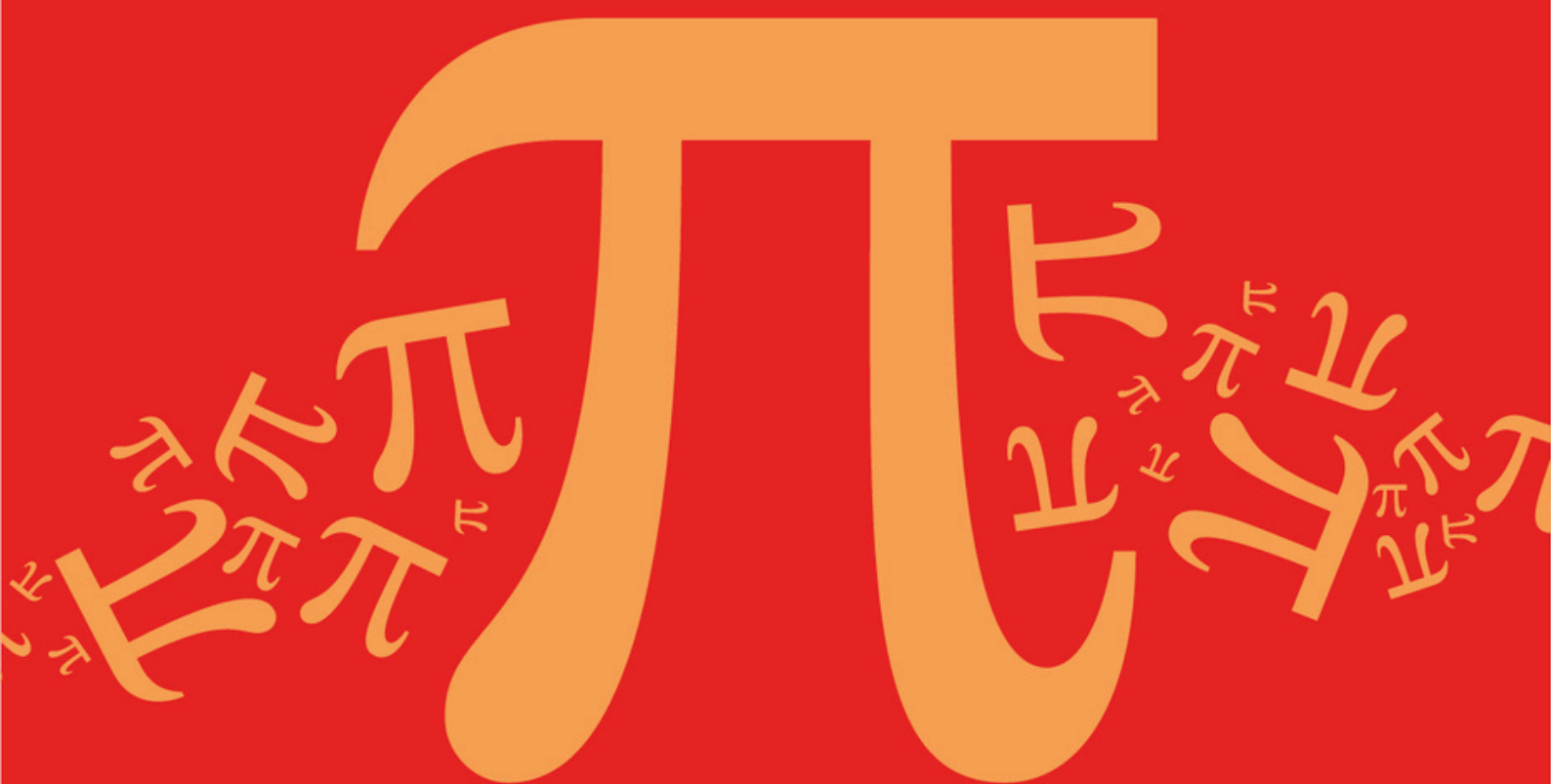
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AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D13: Graph Rationals with Quadratics

Topic Questions

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Question Paper

Topic

D: Further Algebra and Functions

Sub topic

D13: Graph Rationals with Quadratics

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

A curve C_1 has equation $y = f(x)$, where $f(x) = \frac{5x^2 - 12x + 12}{x^2 + 4x - 4}$

The line $y = k$ intersects the curve C_1

- (a) (i) Show that $(k + 3)(k - 1) > 0$

(5)

- (ii) Hence find the coordinates of the stationary point of C_1 that is a maximum point.

(4)

- (b) Show that the curve C_2 , whose equation is $y = \frac{1}{f(x)}$, has no vertical asymptotes.

(2)

- (c) State the equation of the line that is a tangent to both C_1 and C_2 .

(1)

(Total 12 marks)

Q2.

A curve has equation

$$y = \frac{x^2 - 2x + 1}{x^2 - 2x - 3}$$

- (a) Find the equations of the three asymptotes of the curve.

(3)

- (b) (i) Show that if the line $y = k$ intersects the curve then

$$(k - 1)x^2 - 2(k - 1)x - (3k + 1) = 0$$

(1)

- (ii) Given that the equation $(k - 1)x^2 - 2(k - 1)x - (3k + 1) = 0$ has real roots, show that

$$k^2 - k \geq 0$$

(3)

- (iii) Hence show that the curve has only one stationary point and find its coordinates.

(No credit will be given for solutions based on differentiation.)

(4)

- (c) Sketch the curve and its asymptotes.

(3)

(Total 14 marks)

Q3.

The curve C has equation $y = \frac{x}{(x+1)(x-2)}$.

The line L has equation $y = -\frac{1}{2}$.

- (a) Write down the equations of the asymptotes of C . (3)
- (b) The line L intersects the curve C at two points. Find the x -coordinates of these two points. (2)
- (c) Sketch C and L on the same axes.
(You are given that the curve C has no stationary points.) (3)
- (d) Solve the inequality

$$\frac{x}{(x+1)(x-2)} \leq -\frac{1}{2}$$

(3)

(Total 11 marks)

Q4.

A graph has equation

$$y = \frac{x+4}{x^2+9}$$

- (a) Explain why the graph has no vertical asymptote and give the equation of the horizontal asymptote. (2)
- (b) Show that, if the line $y = k$ intersects the graph, the x -coordinates of the points of intersection of the line with the graph must satisfy the equation
- $$kx^2 - x + (9k + 4) = 0$$
- (2)
- (c) Show that this equation has real roots if $-\frac{1}{2} \leq k \leq \frac{1}{18}$. (5)
- (d) Hence find the coordinates of the two stationary points on the curve.
(No credit will be given for methods involving differentiation.) (6)

(Total 15 marks)

Q5.

A curve C has equation $y = \frac{1}{(x-2)^2}$.

- (a) Write down the equations of the asymptotes of the curve C .

(2)

- (b) Sketch the curve C .

(2)

(Total 4 marks)

Q6.

A curve has equation

$$y = \frac{(x-1)(x-3)}{x(x-2)}$$

- (a) (i) Write down the equations of the three asymptotes of this curve.

(3)

- (ii) State the coordinates of the points at which the curve intersects the x -axis.

(1)

- (iii) Sketch the curve.

(You are given that the curve has no stationary points.)

(4)

- (b) Hence, or otherwise, solve the inequality

$$\frac{(x-1)(x-3)}{x(x-2)} < 0$$

(2)

(Total 10 marks)

Q7.

A curve has equation

$$y = \frac{x^2}{(x-1)(x-5)}$$

- (a) Write down the equations of the three asymptotes to the curve.

(3)

- (b) Show that the curve has no point of intersection with the line $y = -1$.

(3)

- (c) (i) Show that, if the curve intersects the line $y = k$, then the x -coordinates of the points of intersection must satisfy the equation

$$(k - 1)x^2 - 6kx + 5k = 0$$

(2)

- (ii) Show that, if this equation has equal roots, then

$$k(4k + 5) = 0$$

(2)

- (d) Hence find the coordinates of the two stationary points on the curve.

(5)

(Total 15 marks)

Q8.

A curve C has equation

$$y = \frac{2}{x(x-4)}$$

- (a) Write down the equations of the three asymptotes of C .

(3)

- (b) The curve C has one stationary point. By considering an appropriate quadratic equation, find the coordinates of this stationary point.

(No credit will be given for solutions based on differentiation.)

(6)

- (c) Sketch the curve C .

(3)

(Total 12 marks)

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Q9.

A curve has equation

$$y = \frac{x}{x^2 - 1}$$

- (a) Write down the equations of the three asymptotes to the curve.

(3)

- (b) Sketch the curve.

(You are given that the curve has no stationary points.)

(4)

- (c) Solve the inequality

$$\frac{x}{x^2 - 1} > 0$$

(3)

(Total 10 marks)

Q10.

A curve C has equation

$$y = \frac{(x+1)(x-3)}{x(x-2)}$$

- (a) (i) Write down the coordinates of the points where C intersects the x -axis. (2)
- (ii) Write down the equations of all the asymptotes of C . (3)
- (b) (i) Show that, if the line $y = k$ intersects C , then

$$(k-1)(k-4) \geq 0$$
 (5)
- (ii) Given that there is only one stationary point on C , find the coordinates of this stationary point.
 (No credit will be given for solutions based on differentiation.) (3)
- (c) Sketch the curve C . (3)
- (Total 16 marks)**

Q11.

The function f is defined by

$$f(x) = \frac{x^2 + 2x + 2}{x^2}$$

- (a) Write down the equations of the two asymptotes to the curve $y = f(x)$. (2)
- (b) By considering the expression $x^2 + 2x + 2$:
- (i) show that the graph of $y = f(x)$ does not intersect the x -axis; (2)
- (ii) find the non-real roots of the equation $f(x) = 0$. (3)
- (c) (i) Show that, if the equation $f(x) = k$ has two equal roots, then

$$4 - 8(1 - k) = 0$$
 (3)
- (ii) Deduce that the graph of $y = f(x)$ has exactly one stationary point and find its coordinates.

(4)
(Total 14 marks)

Q12.

The function f is defined by

$$f(x) = \frac{x^2 + 4x}{x^2 + 9}$$

- (a) (i) The graph of $y = f(x)$ has an asymptote which is parallel to the x -axis.
Find the equation of this asymptote.

(1)

- (ii) Explain why the graph of $y = f(x)$ has no asymptotes parallel to the y -axis.

(2)

- (b) Show that the equation $f(x) = k$ has two equal roots if $9k^2 - 9k - 4 = 0$.

(4)

- (c) Hence find the coordinates of the two stationary points on the graph of $y = f(x)$.

(6)

(Total 13 marks)